BattleBot Challenge

Game Rules and Engine–Bot Interface Specification

January 21, 2026

Abstract

A 20-day multi-agent survival game where five players compete for limited daily water through sealed-bid auctions. This document specifies game rules, mechanics, and the engine–bot communication protocol.

1 Overview

Five players must survive 20 days by acquiring water through daily auctions. Each player has fixed parameters (water requirement and income).

Objective

Survive until Day 20.

2 Player Parameters (Fixed)

Each bot is assigned one player at initialization.

ID	Name	Water Requirement (units/day)	Daily Salary ($/day)
1	Alex	8	70
2	Bob	9	75
3	Cindy	10	100
4	David	11	120
5	Eric	12	120

3 Health Mechanics

3.1 Initial State

- Starting health: **8**
- Maximum health: **10**
- Elimination: health $\leq$ **0**

3.2 Health Updates

Let k = consecutive days without water.

Water Acquired:

$$\text{Health} \leftarrow \min(\text{Health} + 2, 10)$$
$$k \leftarrow 0$$

Water Not Acquired:

$$k \leftarrow k + 1$$
$$\text{Health} \leftarrow \text{Health} - k$$

3.3 Example

Day	**Water?**	k	**Change**	**Health**
10	No	1	-1	$7 \to 6$
11	No	2	-2	$6 \to 4$
12	No	3	-3	$4 \to 1$
13	Yes	0	$+2$	$1 \to 3$
14	No	1	-1	$3 \to 2$

3.4 Elimination

When health ≤ 0:

- Player is permanently eliminated
- Balance reset to \$0
- No further salary

4 Money System

4.1 Starting Balance

Players start with their daily salary amount.

4.2 Daily Salary

Each day, living players receive their fixed salary. Money accumulates.

4.3 Bidding Rules

- Bids must be non-negative integers
- If bid > balance, bid is treated as \$0
- Money deducted **only if player wins water**

5 Water Supply

- Daily supply determined by game engine
- Announced before bidding

6 Auction Mechanism

6.1 Process

1. Each player submits one sealed bid for their full water requirement
2. Bids sorted in descending order
3. Allocate water while: Remaining Supply $\geq$ Player's Requirement
4. All-or-nothing allocation

6.2 Tie-Breaking

Priority to player with smaller water requirement.

6.3 Example

Supply: 30 units
Bids: Cindy($60, 10u), Eric($55, 12u), Alex($50, 8u), Bob($45, 9u), David($40, 11u)
Allocation:

1. Cindy: 10u $\rightarrow$ 20u remain
2. Eric: 12u $\rightarrow$ 8u remain
3. Alex: 8u $\rightarrow$ 0u remain
4. Bob, David: Nothing

7 Communication Protocol

All I/O via standard input/output. Bots output one integer bid per round.

7.1 Round 0: Initialization

Input:

```
3
1 Alex 8 70
2 Bob 9 75
3 Cindy 10 100
4 David 11 120
5 Eric 12 120
```

Format:

- Line 1: Your player ID
- Lines 2–6: `player_id name water_req salary`

Output: None

7.2 Round 1

Input:

```
1
45
1 8 70 0
2 8 75 0
3 8 100 0
4 8 120 0
5 8 120 0
```

Format:

- Line 1: Round number
- Line 2: Water supply
- Lines 3–7: `player_id health balance consecutive_no_water_days`

Output:

```
bid_amount
```

7.3 Round r ($r \geq 2$)

Input:

```
r
52
1 10 120 0
2 6 80 2
3 8 150 0
4 -1 0 0
5 7 95 1
1 45 8
2 30 9
3 50 10
4 0 11
5 35 12
```

Format:

- Line 1: Round number ($2 \leq r \leq 20$)
- Line 2: Water supply
- Lines 3–7: `player_id health balance consecutive_no_water_days`
  - Eliminated: `-1 0 0`
- Lines 8–12: `player_id previous_bid water_requirement`
  - Eliminated player's bid: 0

Output:

```
bid_amount
```

8 Game Termination

Game ends when:

1. Day 20 completes, OR
2. Only one player survives

9 Key Rules Summary

1. Consecutive penalty: k days without water costs k health on day k
2. All-or-nothing allocation
3. Pay only if you win
4. Invalid bids treated as \$0
5. Elimination is permanent

10 Tournament Structure

10.1 Competition Format

- 8 teams compete in the tournament
- Each match consists of 5 teams playing one complete 20-day game
- One tournament round includes all possible 5-team combinations with all possible player class: $\binom{8}{5} \cdot 5! = 6720$ matches
- Ratings are updated after each match round based on match rankings

10.2 Tournament Schedule

- **Checkpoint 1:** TBD (rated or unrated - depends on majority)
- **Checkpoint 2:** TBD (rated or unrated - depends on majority)
- **Finals:** TBD

10.3 Engine and Resources

- Engine code will be made public in due time
- A sample bot template is provided below
- Minor adjustments to the I/O format may occur as the engine develops

11 Sample Bot Template (C++)

```
#include <bits/stdc++.h>
using namespace std;

struct Player {
    int id;
    string name;
    int water_req;
    int salary;
    int health;
    int balance;
    int no_water_days;
};

int main() {
    // Round 0: Initialization
    int my_player_id;
    cin >> my_player_id;

    vector<Player> players(5);
    for (int i = 0; i < 5; i++) {
        cin >> players[i].id >> players[i].name
            >> players[i].water_req >> players[i].salary;
    }
    // Main game loop
    for (int round = 1; round <= 20; round++) {
        int water_supply;
        cin >> round >> water_supply;

        // Read player statuses
        for (int i = 0; i < 5; i++) {
            cin >> players[i].id >> players[i].health
                >> players[i].balance >> players[i].no_water_days;
        }
        // Read previous bids (from round 2 onwards)
        if (round >= 2) {
            vector<int> prev_bids(5);
            vector<int> water_reqs(5);
            for (int i = 0; i < 5; i++) {
                int pid;
                cin >> pid >> prev_bids[i] >> water_reqs[i];
            }
        }

        /* YOUR STRATEGY HERE */
        int my_bid = 0;

        // Output bid
        cout << my_bid << endl;
        cout.flush();
    }

    return 0;
}
```

Good luck!